Open Source Performance Testing *and* Audit Tools *for* Android Apps

Security and performance are key words in the world of mobile apps. There are numerous ways in which a mobile app that is launched without proper testing will flounder and ruin the reputation of the enterprise that launches it, owing to the many threats lurking on the Internet. Performance testing and audits play an important role in the development life cycle of an app. There are many open source tools that can be used for performance audits, some of which are introduced briefly in this article.

Mobile apps face innumerable risks because of their vulnerabilities and it is important to test them on different parameters. Deploying a mobile app without proper and rigorous testing can be dangerous for the vendor. If an e-commerce based online shopping mobile app is launched without testing, hackers and malicious traffic can damage the overall setup and abnormal payments may be made. Often, the apps of e-wallets get exploited and the app users lose money. Such situations hurt the overall reputation and reliability of the company owning the app.

Testing strategies

The performance of mobile apps can be evaluated through a number of testing strategies in a process known as the performance audit. In this process, malicious traffic is intentionally directed to the app so that its behaviour in such an environment can be analysed. There are many types of testing approaches that can be used for a performance audit of the mobile app.

A few key testing strategies for mobile applications are listed below.

Functional testing: This strategy ensures that all the functions, links and buttons on the mobile app are working as desired. Different options and further links are navigated in this process to confirm the overall flow of the application.

Interruption testing: As mobile apps work in the network environment, it is necessary to check their behaviour during assorted interruptions, such as:

- Network coverage
- Incoming and outgoing calls
- Incoming and outgoing SMSs
- Battery removal
- Turning media on/off
- Power cycle
- Transferring data via cable
- Roaming

The app is considered effective only when it's confirmed that all interruptions can be handled without any error, warnings or system-generated exceptions.

Usability testing: This ensures that the application works on different platforms and devices without compatibility issues. The ease with which the application can be used, along with the flexibility in handling different controls and options, is considered in this testing strategy.

Stress testing: The goal of stress testing is to analyse the reliability and stability of the application when it's exposed to different types of abnormal traffic or inputs. Stress testing checks whether the app is getting to any breakpoint on particular inputs. In addition, the modes and points of failure in the app are identified during this test.

Memory leakage testing: As mobile devices have memory constraints (RAM, internal memory, SD card storage,